**EJC 2026 H2 Math Prelim P2**

**Section A: Pure Mathematics [40 marks]**

**1** **Do not use a calculator in answering this question.**

Find the complex numbers $z$ and $w$ which satisfy the simultaneous equations

$$(3+2\mathrm{i})z-3w=3-11\mathrm{i} \quad \text{and} \quad (1-\mathrm{i})z-6\mathrm{i}w=10+8\mathrm{i}.$$ [4]

**2** The curve $C$ has equation $y=\dfrac{\cos x}{\sqrt{1+\sin x}}$ for $0\le x\le\dfrac{\pi}{2}$. The region $R$ is bounded by the curve $C$, the $x$-axis, and the lines $x=0$ and $x=\dfrac{\pi}{6}$.

**(a)** Use the substitution $u=1+\sin x$ to find the exact area of $R$. [4]

The region $S$ is bounded by the curve $C$, the line $y=\dfrac{1}{\sqrt{2}}$, and the $y$-axis.

**(b)** Verify that $y=\dfrac{1}{\sqrt{2}}$ when $x=\dfrac{\pi}{6}$. Hence find the volume of the solid formed when $S$ is rotated through $2\pi$ radians about the $x$-axis. [3]

**3** A curve $C$ has equation $y=\dfrac{5x^2+16x+18}{x^2-4x}$.

**(a)** State the equations of the asymptotes. [2]

**(b)** Using an algebraic method, show that $\dfrac{5x^2+16x+18}{x^2-4x}$ cannot lie between two certain values (to be determined). [4]

**(c)** Hence sketch the curve $C$, indicating clearly any turning point(s). [3]

**4** Referred to the origin $O$, the points $A$ and $B$ have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively. $O$, $A$ and $B$ are **not** collinear. Point $C$ has position vector $2\mathbf{a}+\mathbf{b}$ and point $D$ lies on $AB$ extended such that $AB=BD$.

**(a)** Show that the position vector of $D$ is $2\mathbf{b}-\mathbf{a}$. [2]

**(b)** Given that $OC=OD$, use suitable scalar products to show that $8\mathbf{a}\cdot\mathbf{b}=3|\mathbf{b}|^2-3|\mathbf{a}|^2$. [2]

**(c)** Given also that $|\mathbf{b}|=2|\mathbf{a}|$, show that $\cos\theta=\dfrac{9}{16}$, where $\theta$ is the angle between $\mathbf{a}$ and $\mathbf{b}$. [2]

**(d)** Given further that $\mathbf{a}$ is a unit vector, find the exact area of triangle $OAC$. [4]

**5** **(a)** Given that $x$ is sufficiently small, find the series expansion of $\tan\left(x-\frac{\pi}{3}\right)$ up to and including the term in $x^2$. [4]

**(b)** Hence find an approximation for the value of $\tan\left(-\frac{7\pi}{24}\right)$, giving your answer in the form $a+b\pi+c\pi^2$, where $a$, $b$ and $c$ are exact real constants to be determined. [3]

**(c)** Find the series expansion of $\ln\left|\sec\left(x-\frac{\pi}{3}\right)\right|$ up to and including the term in $x^2$. [3]

## Section B: Probability and Statistics [60 marks]

**6** A fair tetrahedral die has its four faces numbered 1, 2, 3 and 4. A game is played by rolling the die twice, and the number on the face touching the table is recorded each time. Let $X$ be the score obtained from the two rolls, defined as follows:

- If the two numbers recorded are different, $X$ is the absolute difference between them.
- If the two numbers recorded are the same, $X$ is their common value.

**(a)** Find the probability distribution of $X$. [2]

**(b)** Find the expectation and variance of $X$. [2]

To encourage citizens to adopt an active lifestyle, a town council awards fitness cash vouchers to participants. Each participant plays the game three times and is awarded a \$2 cash voucher for every point scored. The participant cashes out the vouchers only after playing the game three times, where a one-time administration fee of \$0.50 is deducted. Let $W$ be the random variable denoting the value, in dollars, of the cash vouchers received by a participant after the administration fee has been deducted.

**(c)** Find E($W$) and Var($W$). [3]

**7** A set of twins, Eugene and Eunice, is involved in a photoshoot with four other sets of twins.

**(a)** Find the number of ways in which ten people can be arranged in a row such that everyone is standing next to his or her twin. [2]

**(b)** The ten people are now arranged in two rows, with 4 people in the front row and 6 people in the back row. Calculate the number of ways this can be done if Eugene and Eunice cannot be next to each other in the same row. [3]

The ten people are now seated at two round tables which can each seat five people. One table is black and the other is white.

**(c)** Find the number of ways in which they can be seated such that no one is seated at the same table as his or her twin. [3]

**8** A meal delivery company, FoodforEu, offers two subscription plans, Basic Plan and Premium Plan, with a 3-day free trial for both plans. Over time, it was found that 76% on the Basic Plan continued their subscription after their trial, while 89% of customers on the Premium Plan continued their subscription after their trial.

In a particular month, 65% of new customers signed up for the Basic Plan 3-day free trial while the rest signed up for the Premium Plan 3-day free trial.

**(a)** Find the probability that

**(i)** a randomly chosen new customer in that month continued his or her subscription after the 3-day free trial, [2]

**(ii)** a randomly chosen new customer in that month chose the Premium Plan given that he or she continued the subscription after the 3-day free trial, [2]

**(iii)** out of 5 randomly chosen new customers in that month who continued their subscription plans, at least one of them signed up for the Premium Plan. [3]

**(b)** Explain why the probability in part **(a)(ii)** is less than 0.5, even though customers on the Premium Plan are more likely to continue their subscription than those on the Basic Plan. [1]

**9** A company manufactures protein bars. The manager claims that the mean mass of the protein bars is 50 grams. A random sample of 40 protein bars is taken, and their masses $x$ grams, are summarised below.

$$\sum(x-50)=-28 \text{ and } \sum(x-50)^2=252.4$$

**(a)** Calculate unbiased estimates of the population mean and variance of the mass of the protein bars. [2]

**(b)** Test, at the 5% level of significance, whether the manager has overstated the mean mass of the protein bars. [4]

**(c)** The manager now wants to determine at 5% level of significance whether the mean mass of a protein bar is different from 50 grams. Without doing a new test, explain how the conclusion will change. [1]

The manufacturer also produces yoghurt tubs and claims that the mean and variance of the yoghurt tubs are 500 grams and 8.24 grams$^2$ respectively. It is assumed that the masses of these yoghurt tubs have a normal distribution. In a study, a random sample of $n$ tubs was taken and the sample mean was 500.50 grams.

**(d)** Find the least value of $n$ such that the null hypothesis, $H_0 : \mu = 500$ would be rejected at the 5% level of significance against the alternative hypothesis, $H_1 : \mu > 500$. [3]

**(e)** State the meaning of the $p$-value of this test in the context of this question. [1]

**10** An automotive analyst is tracking the resale price ($P$ thousand dollars), of used cars over time ($t$ years), after its initial purchase. The following data is collected from several used car listings:

| $t$ | 0.5 | 1 | 2 | 5 | 8 | 12 | 15 | 18 |
|---|---|---|---|---|---|---|---|---|
| $P$ | 98 | 65.2 | 28.5 | 14.8 | 11.2 | 10.4 | 10.1 | 9 |

**(a)** Draw a scatter diagram of the data. Use your diagram to explain whether the relationship between $P$ and $t$ is likely to be well modelled by an equation of the form $P = at + b$, where $a$ and $b$ are constants. [2]

**(b)** The analyst proposes two models (A) and (B) for the relationship between $P$ and $t$, where $a$, $b$, $c$, and $d$ are constants:

$$\text{(A) } P = a + b\left(\frac{1}{t}\right) \qquad \text{(B) } P = c + d\ln t\,.$$

**(i)** By calculating the relevant product moment correlation coefficients, determine whether the relationship between $P$ and $t$ is modelled better by (A) or (B). Explain how you decide which is the better model and state the equation in this case. [5]

**(ii)** Use your equation in part **(b)(i)** to estimate the resale price of a used car 30 years after its initial purchase. Explain whether your estimation is reliable. [2]

**(c)** The analyst wants to publish his report in a Hong Kong car magazine. Re-write your equation from part **(b)(i)** so that it can be used when the price is given in Hong Kong dollars.

[1 Singapore dollar = 6.13 Hong Kong dollars] [1]

The analyst later discovers that the prices listed are inclusive of a $1000 administrative fee.

**(d)** Explain whether the product moment correlation coefficient would differ if the analyst removed this administrative fee from his data. [1]

**11** The masses of mangoes in kg have the distribution $\mathrm{N}\left(0.3, 0.1^2\right)$.

**(a)** Find the probability that two randomly selected mangoes weigh at least 0.5 kg in total. [2]

Mark sells fruit boxes, each of which contain 10 randomly selected blueberries and 2 randomly selected mangoes. Let the number of unblemished blueberries in a randomly selected fruit box be $U$. On average, the probability of a blueberry being unblemished is $p$.

**(b)** State a reason, in the context of the question, why $U$ might **not** be well-modelled by a binomial distribution. [1]

Assume now that $U$ has the distribution $\mathrm{B}(10, p)$. The probability that a randomly selected fruit box contains at least 8 unblemished blueberries is 0.9.

**(c)** Find the value of $p$. [2]

Each fruit box is classified as grade A, B or C according to the following criteria:

| Grade | Criteria |
|---|---|
| A | Fruit boxes that meet <u>both</u> the following criteria:<br>1. Mangoes weigh at least 0.6 kg in total<br>2. The fruit box contains at least 9 unblemished blueberries. |
| B | Fruit boxes that do not meet the criteria for grade A, but meet <u>both</u> the following criteria:<br>1. Mangoes weigh at least 0.5 kg in total<br>2. Contains at least 8 unblemished blueberries. |
| C | All fruit boxes that are not classified as grade A nor B |

**(d)** Show that the probability that a randomly chosen fruit box is classified as grade C is 0.316, correct to 3 significant figures. [2]

**(e)** Hence find the probability that a randomly selected grade C fruit box contains at least 8 unblemished blueberries. [3]

**(f)** The selling prices of a grade A, grade B and grade C fruit box are \$12, \$10 and \$6 respectively. Find the probability that the total selling price of 5 randomly selected fruit boxes is less than \$37. [3]

Mark decides to amend the criteria for grade A boxes as follows:

- **Each** mango in the box weighs at least 0.3 kg **and** the box contains at least 9 unblemished blueberries.

**(g)** State how the percentage of fruit boxes classifed as grade A will change, and explain your answer. [2]